

AER507 Hand-Launched Composite Glider Project

Project Overview

In Fall 2025, our AER507 Materials and Manufacturing team was asked to design, build and fly a hand-launched glider that maximised payload fraction and glide distance. The airframe, excluding payload, could weigh no more than 175 g, the wing area was capped at 108 in², and the tail area was limited to 30 percent of the wing area. The wing had to be a composite part made in the manufacturing lab, so material selection, mould design and layup process were as much a part of the project as the aerodynamics.

The glider carried a 200 g payload at a 175 g airframe mass, a payload fraction of 0.533, and flew repeatable, stable glides of more than 15 m in testing.

Design Objectives

An objective tree and pairwise comparison chart set four priorities that drove every trade-off:

- Payload to weight ratio, since the score depended on it directly.
 - Glide distance and stability, so the aircraft had to be trimmed and repeatable rather than fast.
 - Manufacturability with the materials and tooling available in the lab.
 - Low mass everywhere except where structure was needed.
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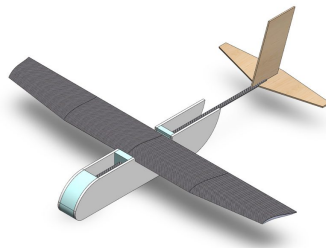


Figure 1. CAD model of the glider: composite wing, display-board fuselage and balsa tail on a carbon fibre boom.

Wing Design and Analysis

A 29.5 in span, 4 in chord planform with 8 degrees of leading-edge taper at the tips gave the full 108 in². The NACA M22 airfoil was selected for its high lift coefficient and gentle stall at low Reynolds numbers, which suits a hand launch where release speed and angle vary from throw to throw.

To size the structure, half the wing was modelled as a cantilever beam carrying the lift for the loaded aircraft at a design load factor of 3. Peak bending stress came to 1.221 MPa. Combined with a target wing density of 0.00156 Mg/m³, this placed the requirement on an Ashby strength versus density chart, where CFRP was the clear choice. The finished hybrid carbon and glass wing weighed 54.9 g, with an estimated modulus of 13.0 GPa and tensile strength of 0.509 GPa, both well above what the load case

needed.

Fuselage and Tail

Several fuselage concepts were compared, including solid and laminated balsa and a 3D-printed ABS nose. The chosen design was a flat display-board skin bonded around rigid foam blocks, which gave a smooth surface, a stiff wing saddle and a slot for removable battery ballast behind the nose. A conventional tail was preferred over a T-tail or V-tail to avoid deep-stall recovery problems and control coupling. Both tail surfaces were cut from 1/8 in balsa and bonded to a carbon fibre boom.

Manufacturing

- **Mould and core.** A two-piece mould was made from a reference foam wing, polished and release-waxed. A 50 g shot of 2.0 lb/ft³ polyurethane foam was poured into the closed mould to form the core, then trimmed and laser-cut with internal pockets to save weight and house fibreglass ribs.
- **Composite layup.** Wet layup with 2.9 oz carbon cloth on the upper surface for stiffness and 2.0 oz fibreglass on the lower surface to save weight. MVS-410 epoxy and MVS-464 hardener were mixed by mass (33.4 g resin, 5.8 g hardener) at roughly twice the fibre mass, and 1.6 oz glass strips were laid along the leading and trailing edges to bond the two skins.
- **Cure and finish.** The core and skins were closed in the mould, bolted and left to cure, then the part was trimmed on a diamond blade saw and sanded to the final profile. The wing was weighed before and after trimming to check the mass budget.
- **Assembly.** The wing mounted on bamboo skewers with rubber bands, which allowed incidence and position to be adjusted between flights and let the wing release on a hard landing. Ballast was moved until the glider balanced at 25 to 33 percent chord.



Figure 2. The finished glider before flight testing.

Results

A three-stage inspection (after part manufacture, after assembly and before flight) checked wing surface quality, skin bonding, tail symmetry and mass balance. In flight testing the glider covered more than 15 m repeatedly with no porpoising or roll-off after release, which showed the trim and alignment were right. The main limitations were sensitivity to launch angle and speed, and the extra mass of epoxy at the glued joints. Alignment jigs and trim tabs were identified as the next improvements.

Skills and Takeaways

- Technical: composite wet layup and mould making, beam-based structural sizing, Ashby material selection, airfoil selection with XFOIL data, SolidWorks CAD, and flight trimming by CG adjustment.
 - Professional: leading a four-person build across design, manufacturing and test, and presenting the design and results to faculty and the cohort at end of term.
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Supplementary materials. The full technical report, including the objective tree, tail sizing rules, wing design calculations and material property derivations, is available here:

[AER507 CompositeGlider FullReport](#). This work was conducted by the AER507 team at **Toronto Metropolitan University: Gaber Soltan, Muhammad Fadhil, Evelyn Hooper and Avalon Snowden** (Fall 2025).